Acute Coronary Syndrome / STEMI

AC-04 | Adult Cardiac

Quick flow

01 Suspected ACS [EMT / AEMT / Paramedic]

Chest discomfort or anginal equivalent • ABCs • symptom-onset time • complete vital signs • SpO_2 • cardiac monitor

02 12-Lead Within 10 Minutes [EMT / AEMT / Paramedic]

Acquire and transmit immediately • repeat for persistent or recurrent symptoms, nondiagnostic tracing, or deterioration

03 Immediate Treatment [EMT / AEMT / Paramedic]

Aspirin 324 mg chewed • oxygen only when indicated • prepare for instability

04 Nitroglycerin Eligible? [EMT / AEMT / Paramedic]

SBP ≥110 • no shock, RV infarction/preload dependence, or prohibited PDE-5 use • 0.4 mg SL every 5 min, max 3

05 STEMI / Occlusion Suspected? [EMT / AEMT / Paramedic]

Transmit ECG • early alert • minimize scene time • choose the fastest appropriate reperfusion pathway

06 Reperfusion Destination [EMT / AEMT / Paramedic]

Hospital with catheterization-lab capability or Claiborne Medical Center for hospital-directed thrombolytic treatment

07 Consider Covenant Health Air [EMT / AEMT / Paramedic]

Activate when appropriate and expected to reduce time to reperfusion-capable care • do not delay ground movement or rendezvous

08 Reassess En Route [EMT / AEMT / Paramedic]

Symptoms • ECG • blood pressure • perfusion • respiratory status • response to treatment

Overview

- Adult acute coronary syndrome pathway for patients 16 years of age and older with chest discomfort, an anginal-equivalent complaint, or ECG findings concerning for acute coronary occlusion.
- Obtain and transmit a diagnostic-quality 12-lead ECG as early as possible, with a goal of within 10 minutes of first medical contact.
- Target scene time is less than 15 minutes unless essential stabilization requires additional time.

EMT Care

Rapid recognition, aspirin, prescribed-nitroglycerin assistance, ECG acquisition, early notification, and time-sensitive transport.

EMT

- Perform the primary assessment, obtain symptom-onset time and focused history, apply the cardiac monitor/AED, obtain SpO_2 and complete vital signs, and place defibrillation pads when instability is present or anticipated.
- Obtain a 12-lead ECG within 10 minutes of first medical contact when equipment and training are available; transmit the tracing immediately without delaying transport.
- Give aspirin 324 mg chewed unless contraindicated.

- Assist with the patient's prescribed nitroglycerin only when SBP is at least 110 mmHg and no contraindication is present.
- Provide oxygen only for SpO_2 below 90%, respiratory distress, shock, or another clinical indication.
- Initiate early destination notification and transport under the Claiborne reperfusion plan.

AEMT Care

All EMT care plus vascular access, agency nitroglycerin, cautious fluid treatment for hypotension, and continued ECG surveillance.

AEMT

- Perform all EMT actions and establish IV access without delaying transport.
- Give nitroglycerin 0.4 mg SL every 5 minutes for continuing ischemic discomfort, to a maximum of three doses, only when SBP is at least 110 mmHg and no contraindication is present.
- For hypotension without pulmonary edema, give isotonic crystalloid in cautious 250 mL aliquots with reassessment after each aliquot.
- Repeat 12-lead ECGs for persistent or recurrent symptoms, a nondiagnostic initial tracing, or clinical deterioration.
- Immediately report hypotension, dysrhythmia, worsening respiratory status, altered mental status, or other evidence of instability.

Paramedic Care

All AEMT care plus ECG interpretation, high-risk occlusion recognition, refractory-pain treatment, instability management, and reperfusion destination decisions.

Paramedic

- Perform all EMT and AEMT actions; interpret serial ECGs and identify STEMI, posterior or right-ventricular involvement, and other patterns concerning for acute coronary occlusion.
- Obtain V4R for suspected right-ventricular involvement or V7-V9 for suspected posterior infarction when indicated and feasible, without delaying transport.
- For persistent significant ischemic discomfort after aspirin and appropriate nitroglycerin, give fentanyl 25-50 mcg IV/IO; repeat as needed to a maximum total dose of 100 mcg.
- Use continuous cardiac, respiratory, SpO_2, and blood-pressure monitoring after fentanyl administration.
- Treat dysrhythmia, acute pulmonary edema, shock, or cardiac arrest under the corresponding linked protocol.
- Choose the fastest appropriate reperfusion pathway, initiate early facility notification, and consider Covenant Health Air when appropriate.

Indications

- Chest discomfort, pressure, heaviness, tightness, or pain concerning for myocardial ischemia.
- Anginal-equivalent symptoms including dyspnea, diaphoresis, nausea, vomiting, syncope, weakness, or jaw, arm, back, shoulder, or epigastric discomfort.
- Diagnostic, dynamic, or evolving ECG changes concerning for STEMI or acute coronary occlusion.
- Unexplained high-risk symptoms in an older adult or a patient with diabetes, known coronary disease, or another significant cardiovascular risk factor.

Contraindications

- No exclusion should prevent appropriate assessment and ECG acquisition when acute coronary syndrome is reasonably suspected.
- Withhold aspirin for true aspirin anaphylaxis, active major bleeding, or strong suspicion of acute aortic dissection.

- Withhold nitroglycerin when SBP is below 110 mmHg, shock or preload dependence is present, right-ventricular infarction is suspected, or prohibited recent PDE-5 inhibitor use is identified.

Assessment | Immediate assessment

- Perform the primary assessment and treat immediate life threats.
- Obtain continuous cardiac monitoring, SpO_2, blood pressure, and two complete vital-sign sets when feasible.
- Determine exact symptom onset, progression, provoking or relieving factors, and associated symptoms.
- Assess for prior myocardial infarction, PCI, CABG, heart failure, dysrhythmia, and implanted cardiac devices.
- Review aspirin already taken, antiplatelet or anticoagulant therapy, active bleeding, medication allergies, and recent PDE-5 inhibitor use.
- Consider other time-critical causes of chest or upper-body discomfort, including aortic dissection, pulmonary embolism, tension pneumothorax, esophageal rupture, and hemorrhage.

Assessment | 12-lead ECG

- Obtain a diagnostic-quality 12-lead ECG as early as possible, with a goal of within 10 minutes of first medical contact.
- Transmit the ECG to the anticipated receiving facility immediately and provide early notification when STEMI or acute coronary occlusion is suspected.
- Repeat the ECG for persistent or recurrent symptoms, a nondiagnostic initial tracing, or clinical deterioration.
- Obtain V4R for suspected right-ventricular involvement or V7-V9 for suspected posterior infarction when indicated and feasible.
- Do not delay transport solely for ECG transmission, repeat ECGs, or additional lead placement.

Assessment | STEMI identification

- Use the clinical presentation together with ST elevation in at least two anatomically contiguous leads.
- Measure ST elevation at the J point relative to the isoelectric baseline.
- In leads other than V2-V3: at least 1 mm of ST elevation in two contiguous leads.
- In V2-V3 for women: at least 1.5 mm.
- In V2-V3 for men age 40 or older: at least 2 mm.
- In V2-V3 for men younger than 40: at least 2.5 mm.
- The monitor's automated interpretation does not independently confirm or exclude STEMI.

Assessment | Contiguous lead groups

- Inferior: II, III, aVF.
- Lateral: I, aVL, V5, V6.
- Anterior/septal: V1-V4.
- Posterior: V7-V9.
- Right ventricular: V3R-V4R.

Assessment | STEMI equivalents and high-risk occlusion patterns

- Posterior MI: horizontal ST depression in V1-V3 with prominent R waves or upright T waves in the appropriate clinical setting; obtain V7-V9 when feasible.
- Right-ventricular involvement: in inferior STEMI, obtain right-sided leads when feasible; ST elevation in V4R supports RV involvement.
- New or presumed-new left bundle branch block alone is not an automatic STEMI criterion.

- In left bundle branch block or paced rhythm, immediately transmit tracings with concordant ST elevation, concordant ST depression in V1-V3, excessive discordance, dynamic ischemia, or another pattern strongly concerning for acute occlusion.
- Other high-risk patterns include diffuse ST depression with ST elevation in aVR, de Winter-type changes, hyperacute T waves, and rapidly progressive ST-segment changes.

Treatment steps

- Administer aspirin 324 mg chewed as soon as acute coronary syndrome is suspected unless contraindicated. If the patient has already taken aspirin, give enough to complete a 324 mg loading dose.
- Provide oxygen only for SpO_2 below 90%, respiratory distress, shock, or another clinical indication; do not routinely administer oxygen when SpO_2 is at least 90%.
- Assist with prescribed nitroglycerin at the EMT level or administer agency nitroglycerin at the AEMT/Paramedic level: 0.4 mg SL every 5 minutes as needed, maximum three doses.
- Withhold nitroglycerin when SBP is below 110 mmHg, shock or preload dependence is present, right-ventricular infarction is suspected, or the patient used avanafil within 12 hours, sildenafil or vardenafil within 24 hours, or tadalafil within 48 hours.
- Establish IV access without delaying transport. Do not routinely administer fluids; for hypotension without pulmonary edema, use cautious 250 mL isotonic-crystalloid aliquots with reassessment.
- For persistent significant ischemic discomfort after aspirin and appropriate nitroglycerin, a paramedic may give fentanyl 25-50 mcg IV/IO and repeat as needed to a maximum total dose of 100 mcg with continuous monitoring.
- Immediately notify the anticipated receiving facility and transmit the ECG when STEMI or acute coronary occlusion is suspected.
- Transport to a hospital with cardiac catheterization laboratory capability or to Claiborne Medical Center for hospital-directed thrombolytic treatment under the Claiborne destination plan.
- Consider Covenant Health Air when appropriate and reasonably expected to reduce time to reperfusion-capable care.
- Air activation must not delay assessment, aspirin, ECG acquisition, stabilization, or movement toward ground transport or rendezvous.
- Treat dysrhythmia, acute pulmonary edema, shock, or cardiac arrest under the corresponding linked protocol.
- Target scene time is less than 15 minutes. Do not delay transport for repeat ECGs, IV access, additional leads, or nonessential procedures.

Medications

Aspirin

324 mg PO chewed

- EMT, AEMT, and Paramedic standing order for suspected acute coronary syndrome.
- If aspirin was taken before EMS arrival, give enough to complete a total loading dose of 324 mg.
- Withhold for true aspirin anaphylaxis, active major bleeding, or strong suspicion of acute aortic dissection.

Nitroglycerin

0.4 mg SL every 5 minutes as needed; maximum 3 doses

- EMT may assist with the patient's prescribed nitroglycerin; AEMT and Paramedic may administer agency nitroglycerin.
- Withhold when SBP is below 110 mmHg, shock or preload dependence is present, right-ventricular infarction is suspected, or prohibited recent PDE-5 inhibitor use is identified.
- Reassess symptoms, blood pressure, and perfusion before and after every dose.

Fentanyl

25-50 mcg IV/IO; repeat as needed to a maximum total dose of 100 mcg

- Paramedic standing order for persistent significant ischemic discomfort after aspirin and appropriate nitroglycerin.
- Use continuous cardiac, respiratory, SpO_2, and blood-pressure monitoring.
- Avoid routine opioid administration; opioids may delay absorption of oral antiplatelet medications.

Isotonic crystalloid

250 mL IV/IO aliquots with reassessment

- AEMT and Paramedic for hypotension without pulmonary edema.
- Do not administer routine fluid boluses to a normotensive patient.

Warnings / precautions

- A normal or nondiagnostic initial ECG does not exclude acute myocardial infarction.
- Do not use a new or presumed-new left bundle branch block alone as an automatic STEMI criterion.
- Withhold nitroglycerin for SBP below 110 mmHg, shock, suspected right-ventricular infarction or preload dependence, and prohibited recent PDE-5 inhibitor use.
- Strong suspicion of acute aortic dissection is a reason to withhold aspirin and nitroglycerin and pursue immediate time-critical transport.
- Do not delay transport while waiting for air-medical arrival when ground transport or rendezvous is faster.
- Claiborne EMS does not administer prehospital thrombolytics, heparin, or a P2Y12 inhibitor under AC-04.
- Transport to Claiborne Medical Center for thrombolytic treatment means hospital evaluation and hospital-directed treatment; EMS does not independently determine thrombolytic eligibility.

Clinical pearls

- Transmit the first ECG as soon as possible, even when the automated interpretation is nondiagnostic.
- Early notification allows the receiving team to review the tracing and prepare the catheterization laboratory or thrombolytic pathway.
- Serial ECGs are particularly valuable when symptoms continue or the initial tracing is nondiagnostic.
- Borderline ST elevation with strong reciprocal changes may still represent acute coronary occlusion.
- Older adults, women, and patients with diabetes may present without classic chest pain.

Special populations | Atypical presentations

- Consider acute coronary syndrome in unexplained dyspnea, weakness, diaphoresis, nausea, syncope, or jaw, arm, shoulder, back, or epigastric discomfort.
- Maintain a low threshold for ECG acquisition in older adults and patients with diabetes or known coronary disease.

Special populations | Right-ventricular infarction

- Inferior STEMI with hypotension, clear lungs, jugular venous distention, or disproportionate preload dependence should prompt right-sided lead acquisition when feasible.
- Withhold nitroglycerin when right-ventricular infarction or preload dependence is suspected.

Related pathways / contacts

AC-02 Bradycardia With a Pulse - Open for symptomatic bradycardia or poor perfusion.

AC-05 Acute Pulmonary Edema - Open for cardiogenic pulmonary edema or decompensated heart failure.

AC-03 Adult Cardiac Arrest - Open immediately if the patient becomes pulseless.

Source and references

American College of Cardiology/American Heart Association. 2025 Guideline for the Management of Patients With Acute Coronary Syndromes. | American Heart Association and American Red Cross. 2024 Guidelines for First Aid. | Tennessee Emergency Medical Services Board. Tennessee EMS Protocol Guidelines, September 2025. | Claiborne EMS medication formulary and reperfusion destination plan. | Claiborne Medical Center hospital thrombolytic-treatment pathway. | Covenant Health Air.

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